Predicting House Prices: Linear Regression (Ames, Iowa)

A multiple linear regression model that predicts house sale prices from **living area** and **garage area**. HyperionDev Linear Regression capstone project.

Data: Ames Housing dataset: 2,930 house sales in Ames, Iowa (2006–2010), 14 variables
Tools: Python, pandas, NumPy, Matplotlib, seaborn, scikit-learn

What I did

- Checked the data for missing values (none) and data types.
- Explored it with histograms and a correlation heatmap.
- Chose `Gr_Liv_Area` (living area) and `Garage_Area` as predictors. Both are strongly correlated with sale price (0.71 and 0.64).
- Plotted each predictor against sale price to check for a linear relationship.
- Split the data 75% train / 25% test and trained a multiple linear regression model.
- Evaluated the model with MSE and RMSE, and plotted predicted against actual prices.
- Interpreted the coefficients against the median sale price.

Results

Measure	Value
Living area coefficient	$78.69 per sq ft
Garage area coefficient	$141.15 per sq ft
Intercept	–$5,059
RMSE	$51,326
Median sale price	$160,000

Against the median house, an extra 100 sq ft of living space adds about **$7,869**, and an extra 100 sq ft of garage adds about **$14,115**.

Conclusion

Living area and garage size are both strong drivers of price. The model predicts typical, lower-priced homes well, but errors grow for large, expensive houses. Adding features such as basement size or year built would likely improve accuracy.

Run it

```
pip install pandas numpy matplotlib seaborn scikit-learn jupyter
jupyter notebook Linear_Regression_Ames.ipynb
```